

FROM PROPERTY SEARCH TO GOVERNED DECISION INTELLIGENCE

A 50-AGENT HYBRID SWARM FOR RESIDENTIAL REAL ESTATE

Designing a Conversational, Explainable and Resilient Architecture for More Intelligent Property Purchases

A pedagogical research paper on how heterogeneous multi-agent swarms, deterministic financial engines, language-model reasoning, Pareto analysis and explicit governance can be combined to transform an ambiguous buyer conversation into a defensible opportunity set and an executable Colab application.

RESEARCH PROPOSITION

Residential property choice is not fundamentally a database-filtering problem. It is a multi-objective decision under uncertainty in which the buyer's objective function is only partially observed, financial feasibility changes with market conditions, qualitative and quantitative attributes interact, and reasonable specialists may disagree. This paper investigates whether a heterogeneous swarm of fifty artificial specialists, embedded in a hybrid architecture that assigns semantic interpretation to a language model and exact computation to deterministic code, can produce a richer, more resilient and more explainable decision process than a monolithic recommendation engine. The contribution is methodological and pedagogical: the prototype is not offered as a production real-estate advisory system, but as an executable demonstration of how distributed artificial judgment can be organized, governed and converted into a usable application.

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Educational research prototype. The synthetic data and resulting recommendations are not appraisals, brokerage services, mortgage advice, investment advice, legal advice or tax advice. A production system would require validated data rights, jurisdiction-specific controls, professional oversight, security, privacy, fair-housing review and empirical evaluation.

¹The companion Google Colab notebook, `NB_50_AGENT_REAL_ESTATE_SWARM_CLAUDE_SONNET_5_10_CELLS.ipynb`, is available here: [Open the companion Google Colab notebook](#). It contains the executable synthetic market, conversational Buyer Mandate, 50-agent specialist architecture, deterministic governance, Pareto analysis, Claude Sonnet 5 deliberation layer, robustness experiment, provenance record and Gradio demonstration application on which this paper is based.

Abstract

Buying a home is often represented digitally as a search problem: choose a location, set a maximum price, specify bedrooms and bathrooms, and inspect the resulting listings. This representation is computationally convenient but economically incomplete. Residential property choice is a high-dimensional decision under uncertainty involving financing, carrying costs, household needs, schools, commute, neighborhood quality, architectural preference, condition, renovation burden, insurance, climate exposure, liquidity, resale value and an uncertain holding horizon. More fundamentally, the buyer's utility function is not fully known at the start of the process. Stated preferences are incomplete and may be inconsistent with subsequent behavior.

This paper develops and studies a pedagogical prototype in which a heterogeneous swarm of fifty artificial specialists acts as a digital buyer's committee. The architecture is deliberately hybrid. A language model is used for semantic tasks: interpreting an unstructured buyer conversation and synthesizing trade-offs among leading candidates. Deterministic code is used for mortgage mathematics, carrying-cost estimation, feature engineering, specialist scoring, hard-constraint enforcement, Pareto analysis, sensitivity testing and provenance. Fifty agents are organized into Finance, Investment, Family, Location, Lifestyle, Risk and Governance coalitions. Their disagreement is retained as information rather than eliminated as noise. Hard constraints cannot be voted away by an attractive narrative. A Pareto frontier prevents the ranking mechanism from hiding efficient alternatives that embody different trade-offs. An interest-rate experiment asks whether the recommendation survives a material change in financing assumptions.

The experiment is implemented in a Google Colab notebook using a synthetic but economically structured universe of 1,200 properties and culminates in a conversational Gradio application. The application demonstrates an end-to-end path from natural-language interview to formal Buyer Mandate, governed opportunity set, coalition comparison, narrative synthesis and model provenance. The central research claim is not that fifty agents necessarily outperform every simpler recommender. Rather, the prototype shows that swarm intelligence can be organized as an auditable institutional architecture: diverse specialists, explicit evidence, bounded mandates, preserved dissent, deterministic controls and a final synthesis. This provides a useful research platform for studying when distributed artificial judgment improves robustness, explainability and decision quality in complex consumer finance. The paper concludes by proposing an empirical agenda centered on ablation tests, subswarm convergence, revealed-preference learning, calibration, fairness, live-data integration and controlled comparisons with monolithic models and conventional property portals.

Image omitted from the lightweight GitHub edition.

Figure 1: The real-estate decision as a swarm problem — The opening notebook infographic reframes property purchase from a simple listing search into a multidimensional decision involving finance, family, location, lifestyle, investment value, risk and governance.

Introduction: Real Estate Search Is Not Really a Search Problem

Digital real-estate platforms have made property discovery extraordinarily efficient. A buyer can specify geography, price, bedrooms, bathrooms and property type and receive hundreds of candidate listings in seconds. Yet the improvement in discovery should not be confused with a complete solution to the decision problem. A home is simultaneously a consumption good, a durable asset, a leveraged balance-sheet position, a location choice and, for many households, the largest single financial commitment they will make. The attributes that matter are numerous, correlated, uncertain and evaluated over different horizons.

The familiar search interface suppresses much of this complexity. Purchase price is visible, but financing conditions determine the monthly burden. Square footage is visible, but usable space, household composition and future family changes determine whether the space is valuable. School quality, commute, safety, walkability and amenities interact with one another and with price. Property age can create architectural appeal and neighborhood character while simultaneously increasing maintenance and renovation risk. A highly desirable property may be financially fragile under a different mortgage rate. A home that fits today's household perfectly may have weak resale liquidity if the buyer must move unexpectedly.

The decision is therefore naturally represented as a multi-criteria problem rather than a simple retrieval task. Multi-criteria decision analysis has long emphasized that complex choices require explicit treatment of objectives, trade-offs and value judgments (Keeney and Raiffa, 1976; Saaty, 1980). The real-estate literature similarly recognizes that property values are bundles of characteristics rather than a single primitive attribute (Rosen, 1974). What generative artificial intelligence changes is not the existence of those trade-offs. It changes our ability to build a conversational interface around them and to distribute the analysis across artificial specialists.

A second difficulty is even more important: the buyer's preferences are only partially observed. The client may say that schools dominate every other concern, then repeatedly save homes in more walkable neighborhoods with somewhat weaker schools. A buyer may insist on modern architecture but accept an older property when the lot, commute and neighborhood compensate for age. A nominal purchase-price ceiling may prove less important than monthly carrying cost. In the language of decision theory, the system does not begin with a complete and stable utility function. It must infer a provisional mandate from language and then, in a more mature implementation, update that mandate from behavior.

This motivates the research question:

Can a heterogeneous society of artificial specialists, embedded in a governed hybrid architecture, help transform an ambiguous real-estate search into a more intelligent, resilient and explainable purchasing process?

The word *help* is deliberate. The prototype does not attempt to automate the buyer out of the decision. It attempts to improve the quality of the decision environment: formalize the mandate, widen the set of perspectives, expose trade-offs, challenge attractive but fragile candidates, preserve dissent, test sensitivity to assumptions and make the path from evidence to recommendation inspectable.

The resulting architecture resembles an institutional investment committee more than a conventional recommender. Fifty specialists observe the same opportunity through different lenses. Finance agents examine affordability, rates, taxes, insurance and maintenance. Investment agents examine resale, appreciation, rental optionality and negotiation signals. Family agents emphasize schools, safety and space. Location agents focus on commute, transit, walkability and amenities. Lifestyle agents consider architecture, condition and subjective fit. Risk agents search for climate, age, renovation, liquidity and budget fragility. Governance agents ask whether the system is violating the mandate or relying on weak evidence.

The central methodological choice is hybridization. Large language models are powerful semantic systems, but they are not the correct tool for every subproblem. Mortgage amortization has an exact formula. Constraint satisfaction is Boolean. Pareto efficiency is mathematical. Provenance should be recorded mechanically. Conversely, natural-language interviewing, mandate interpretation and explanation of competing trade-offs are precisely the kinds of tasks for which language models can be valuable. The architecture therefore assigns different forms of cognition to the mechanisms best suited to them.

This separation is important for both performance and governance. A system that asks one language model to ingest raw listings, infer preferences, calculate financing, rank opportunities and explain the result creates an unnecessarily large epistemic surface. Errors in arithmetic, interpretation, data, preferences and narrative become difficult to distinguish. The hybrid swarm instead decomposes the problem into inspectable stages.

Image omitted from the lightweight GitHub edition.

Figure 2: The first architectural move: separating forms of intelligence — The notebook establishes a resilient environment in which deterministic computation, language interpretation and provenance are separate components rather than an undifferentiated model call.

Research Objective and Contribution

The research objective is exploratory rather than confirmatory. We ask whether swarm and hybrid architectures provide a useful design pattern for residential real-estate decision support. The prototype is intended to be robust enough to expose architectural strengths and weaknesses, but it is not presented as evidence that a particular algorithm improves realized household welfare in live markets.

The paper makes five contributions.

First, it reframes property discovery as *decision intelligence*. The relevant output is not merely a list of matching homes but a governed opportunity set whose members can be explained relative to the buyer's objectives.

Second, it treats heterogeneity as an architectural resource. A swarm is useful only if its members contribute genuinely different lenses. Fifty copies of the same prompt are not a meaningful society of experts. The prototype therefore creates explicit specialist roles and coalition structures.

Third, it introduces a hybrid allocation of cognition. Semantic interpretation and committee synthesis are assigned to a language model; exact calculations, validation, constraints, ranking mechanics and robustness checks remain deterministic.

Fourth, it treats disagreement as evidence. Dispersion among agents is retained and displayed. A high average score with large dissent means something different from a high score with broad agreement.

Fifth, it closes the loop into an application. The research architecture is not left as a conceptual diagram. The notebook ends with a working conversational Colab app. This matters pedagogically because an executable interface forces the architecture to become operational: inputs, state, calculations, outputs and provenance must actually connect.

WHAT THE PAPER DOES NOT CLAIM

The experiment does not establish that fifty agents are an optimal swarm size, that the synthetic scoring functions are calibrated to household welfare, that Claude Sonnet 5 is uniquely suited to the task, or that the prototype should be used for live property purchases without additional validation. The notebook is a controlled laboratory for architectural reasoning. Its value lies in demonstrating what can be made explicit, testable and governable once a complex decision is decomposed into a society of bounded specialists.

Conceptual Foundations

Multi-criteria choice and the hidden objective function

Let a property j be represented by an attribute vector

$$\mathbf{x}_j = (p_j, s_j, b_j, q_j, c_j, \ell_j, r_j, \dots),$$

where the components may include price, size, bedrooms, school quality, commute, location attributes, risk and liquidity. A conventional filter asks whether selected components satisfy thresholds. A richer decision model asks how the buyer values the complete bundle.

If the buyer's utility function were known, one might write

$$U_j = U(\mathbf{x}_j; \boldsymbol{\theta}),$$

where $\boldsymbol{\theta}$ contains the buyer's preference parameters. The practical problem is that $\boldsymbol{\theta}$ is not observed. The interview provides noisy information about it. Subsequent saves, rejections, tours and reconsiderations can provide additional revealed-preference evidence.

The architecture therefore distinguishes a *Buyer Mandate* from an unknowable "true" utility function. The mandate is a structured, revisable representation of what the system currently believes matters. It contains both hard

constraints and soft priorities. This is a practical application of bounded rationality: complex decisions are made with incomplete information, limited attention and imperfectly articulated objectives (Simon, 1955, 1957).

Why a swarm?

The classical argument for distributed intelligence is not that groups are magically wise. Collective performance depends on diversity, independence, aggregation and the structure of the task (Condorcet, 1785; Page, 2007). Correlated error can make a large group confidently wrong. Groupthink can suppress dissent (Janis, 1972). Multi-agent systems research similarly emphasizes that useful collective behavior emerges from the organization of interacting agents, not merely from agent count (Wooldridge, 2009).

This motivates a strong design principle:

Agent multiplicity without mandate heterogeneity is not swarm intelligence; it is repeated sampling from a similar cognitive process.

The notebook operationalizes diversity through specialist mandates and coalition structure. The objective is not to manufacture personality. It is to create different loss functions: affordability and architecture should not be evaluated as if they were the same question.

Hybrid intelligence

The architecture also reflects a distinction between probabilistic and deterministic competence. Language models are useful where the state is expressed in natural language, where concepts must be mapped into structured fields, and where competing considerations require synthesis. They are less appropriate where an exact financial formula, transparent rule or reproducible optimization is available.

A hybrid architecture can therefore be represented as

$$\mathcal{D} = \mathcal{G} [\mathcal{A} (\mathcal{F} (\mathcal{S} (\mathcal{M}(C))))],$$

where C is the buyer conversation, $\mathcal{M}$ constructs the mandate, $\mathcal{S}$ searches and scores the opportunity space, $\mathcal{F}$ preserves efficient alternatives and disagreement, $\mathcal{A}$ synthesizes the committee evidence, and $\mathcal{G}$ imposes governance. The important feature is that these operators need not all be implemented by the same model.

Experimental Design: A Synthetic Real-Estate Laboratory

The experiment requires a market rich enough to generate real trade-offs but controlled enough to be reproducible. The notebook therefore constructs 1,200 synthetic properties across eight synthetic neighborhoods. Neighborhoods differ in latent prestige, schools, walkability, family suitability, culture, resale characteristics, transit and amenities. Property-level variables are then generated conditionally so that the database contains economically plausible correlations rather than independent random numbers.

The synthetic records include price, property type, style, bedrooms, bathrooms, square footage, lot size, age, condition, tax rate, HOA cost, commute, school score, walkability, family suitability, culture, amenities, safety, noise, renovation estimate, days on market, price-cut indicator, appreciation estimate, rental yield, resale liquidity and climate risk.

This is not an attempt to simulate a particular metropolitan area. It is a laboratory. The objective is to create enough structure that specialists can reasonably disagree. A family coalition should sometimes prefer a different home from an investment coalition. A finance coalition should react to a property whose sticker price appears acceptable but whose carrying cost is not. A lifestyle coalition should occasionally value a home that governance rejects.

Image omitted from the lightweight GitHub edition.

Figure 3: A synthetic but economically structured property market — The second notebook stage creates a reproducible opportunity universe with correlated financial, physical, neighborhood, family, investment and risk attributes.

The data architecture is deliberately replaceable. The synthetic source implements the same normalized schema that a production adapter would populate from authorized listing feeds, public property records, permitted school

and geospatial sources, mortgage rates, insurance information and other lawful datasets. The intelligence layer is therefore separated from any particular portal.

This separation matters commercially and scientifically. A decision system should not confuse the availability of one data vendor with the architecture of intelligence. The market adapter can change while the mandate, swarm, governance and explanation layers remain stable.

From Conversation to a Formal Buyer Mandate

A conventional search form requires the buyer to translate life circumstances into fields before the system can help. The notebook reverses this burden. The client begins with an ordinary conversation.

A buyer may say:

We are a family of four. Schools and safety matter a great deal. I work downtown and would prefer a commute below roughly thirty-five minutes. We like modern homes, want enough space to stay for several years, and care about resale because a future relocation is possible. We can renovate a little, but we do not want a major project.

The language model does not immediately recommend a house. Its first responsibility is narrower: construct a structured mandate.

Let the mandate be

$$M = (H, P, \phi),$$

where H is the set of hard constraints, P the vector of soft priorities, and ϕ the financing and horizon assumptions. Hard constraints can include budget, minimum bedrooms, minimum bathrooms and minimum square footage. Soft priorities can include schools, safety, commute, walkability, resale, condition, style, space, investment value, culture, climate resilience and renovation tolerance. Financing assumptions include down payment, mortgage rate, term and acceptable monthly burden.

The distinction between H and P is essential. A recommendation system becomes unreliable if it treats every stated preference as equally rigid. Conversely, it becomes untrustworthy if it allows an attractive score to override a genuine constraint.

Image omitted from the lightweight GitHub edition.

Figure 4: Conversation becomes a Buyer Mandate — The semantic layer converts natural language into a machine-readable decision contract before any property is recommended, making interpretation inspectable and correctable.

The notebook contains a deterministic fallback parser so the architecture remains executable when the live language-model API is unavailable. This is not intended to match the semantic richness of a language model. It demonstrates a resilience principle: failure of one probabilistic component should not necessarily collapse the complete application.

Financial Engineering Before Artificial Judgment

Property price is not the same as affordability. The notebook therefore constructs an explicit financing layer before the swarm evaluates opportunities.

For purchase price P , down-payment fraction d , annual mortgage rate r , and term of T years, the financed principal is

$$L = P(1 - d).$$

With monthly rate $i = r/12$ and $n = 12T$ payments, the monthly mortgage payment is

$$PMT = L \frac{i(1+i)^n}{(1+i)^n - 1}.$$

The prototype then adds monthly property tax, HOA charges, an insurance proxy and maintenance to obtain an estimated all-in monthly carrying cost:

$$C_j^{all} = PMT_j + Tax_j + HOA_j + Ins_j + Maint_j.$$

This is pedagogically important because it prevents the language model from becoming a numerical oracle. The model can discuss the implications of a carrying cost, but it does not need to improvise the arithmetic.

The same principle extends to future value. Under a synthetic annual appreciation assumption g_j and holding horizon h ,

$$FV_j = P_j(1 + g_j)^h.$$

This is not a forecast in the empirical sense; it is a scenario variable used by investment-oriented agents.

Image omitted from the lightweight GitHub edition.

Figure 5: The financial engine and the society of specialists — Exact financing calculations are completed before specialist judgment. The resulting economic state is then examined by fifty agents with differentiated mandates.

The 50-Agent Society

The swarm contains seven coalitions and fifty named specialists. The purpose of the coalition layer is not merely organizational convenience. It creates an intermediate level between individual opinion and global consensus.

Table 1: Illustrative coalition structure of the 50-agent real-estate swarm

Coalition	Agents	Representative responsibilities
Finance	8	Affordability, mortgage structure, interest-rate risk, taxes, insurance, maintenance, renovation budget, total cost of ownership.
Investment	8	Resale liquidity, appreciation, rental optionality, downside risk, negotiation opportunity, market timing, price per square foot, neighborhood momentum.
Family	8	Schools, safety, family space, bedrooms, bathrooms, outdoor space, child friendliness, multigenerational flexibility.
Location	8	Commute, transit, walkability, amenities, culture, noise, accessibility, neighborhood fit.
Lifestyle	8	Architecture, style, condition, move-in readiness, privacy, entertaining, work-from-home suitability, emotional fit.
Risk	6	Climate exposure, property-age risk, liquidity stress, budget stress, renovation execution, concentration risk.
Governance	4	Data quality, constraint auditing, devil's advocacy, buyer advocacy.

The specialists are not independent models in the strongest computational sense. In the prototype, much of individual scoring is deterministic. What makes them agent-like is the explicit decomposition of the objective into bounded specialist mandates and the preservation of their separate outputs. This is a useful reminder that an “agent” need not always mean an unconstrained language model executing arbitrary tools. Agency can be narrow, role-specific and governed.

Specialist scores

Raw property variables are transformed into normalized features $f_{jk} \in [0, 1]$, where j indexes properties and k indexes decision dimensions. For agent a , a simplified score can be represented as

$$s_{aj} = \frac{\sum_k w_{ak}(M) f_{jk}}{\sum_k w_{ak}(M)}.$$

The weights $w_{ak}(M)$ depend on the agent's coalition, specialty and the buyer mandate. A Schools agent receives a stronger school-quality loading than a general Family agent. An Affordability agent emphasizes purchase-price and monthly-cost fit. The buyer's stated priorities can increase the relevance of a dimension without erasing the specialist's identity.

This creates a useful tension. The swarm listens to the buyer, but the buyer does not fully dictate how every specialist thinks. If climate risk receives low stated priority, the Climate Risk agent does not cease to exist. It becomes a dissenting perspective that the buyer may ultimately choose to discount.

Coalition aggregation and disagreement

For coalition c with agents A_c , the coalition score for property j is

$$S_{cj} = \frac{1}{|A_c|} \sum_{a \in A_c} s_{aj}.$$

A global consensus score can then be constructed from coalition weights ω_c :

$$S_j = \sum_c \omega_c S_{cj} - \lambda V_j - \gamma I_j,$$

where V_j is a constraint-violation penalty and I_j indicates failure of a hard governance gate.

The prototype also records dispersion across agents,

$$D_j = \text{sd}(s_{1j}, \dots, s_{50j}),$$

which serves as a simple measure of dissent.

A property with $S_j = 0.82$ and low D_j is a different committee outcome from a property with the same mean and high D_j . The second deserves more scrutiny because its apparent consensus is hiding disagreement.

Image omitted from the lightweight GitHub edition.

Figure 6: From specialist scores to governed collective ranking — The swarm aggregates bounded specialist judgments while preserving coalition structure, dissent and independent governance constraints.

Governance: Why the Swarm Is Not a Democracy

A pure voting system would be inappropriate for property advice. Suppose forty-six agents favor a beautiful property with strong schools and resale prospects, but the home exceeds the buyer's hard budget. The correct response is not to declare a landslide victory. A constraint is not a minority opinion.

The notebook therefore implements a governance gate outside the popularity mechanism. Budget, minimum bedrooms, bathrooms and square footage are treated as hard constraints in the baseline experiment. Monthly burden and commute are monitored as important feasibility conditions that can be modeled more softly depending on the mandate.

This yields a broader principle for agentic systems:

Governance should constrain collective intelligence; it should not be reducible to collective intelligence.

The same principle appears in institutional finance. An investment committee may have broad discretion, but regulatory limits, risk limits and legal restrictions are not simply additional votes. In an AI architecture, this means that certain rules should remain deterministic and external to generative persuasion.

The governance logic is already embedded in the specialist-scoring stage shown in Figure 6. Once infeasible candidates have been identified and penalized, the notebook moves to a different problem: how to preserve attractive alternatives when no single property is best on every dimension.

Image omitted from the lightweight GitHub edition.

Figure 7: Pareto frontier, candidate packets and advocate-critic challenge — Code Unit 6 moves beyond a single consensus score. It identifies non-dominated alternatives, constructs compact evidence packets for the leading candidates, and makes disagreement operational by showing which specialists become the strongest advocates and critics of each opportunity.

Beyond a Single Score: Pareto Efficiency and Adversarial Challenge

A scalar ranking is useful for navigation but dangerous as a representation of a genuinely multi-objective choice. Two properties can embody different efficient trade-offs. One may dominate on schools and space; another on walkability and resale; a third on affordability and renovation potential.

Let $\mathbf{z}_j$ denote the vector of coalition-level or objective-level scores for property j . Property j is Pareto dominated if there exists another property k such that

$$z_{km} \geq z_{jm} \quad \forall m,$$

with strict inequality for at least one dimension. The Pareto frontier is the set of non-dominated candidates.

This mechanism prevents the global weighting system from erasing alternatives that would be optimal under a plausible change in buyer priorities. In real estate, where preference weights are uncertain, this is particularly valuable.

The notebook also constructs compact evidence packets for leading candidates. Rather than giving the language model the entire 1,200-row universe, the deterministic system provides factual packets containing property characteristics, financial burden, coalition scores, consensus, dissent, violations and Pareto status. The language model reasons over a bounded evidence environment.

The advocate-versus-critic mechanism is equally important. For a leading property, the system can identify the strongest supporting specialists and the strongest skeptical specialists. This is not merely an explanatory flourish. It is an attempt to institutionalize challenge.

In organizational decision making, the quality of a recommendation often depends on whether objections can survive long enough to be considered. A swarm architecture can make dissent computationally persistent. The critic does not need social courage; the system simply preserves the score.

Claude Sonnet 5 as Deliberator, Not Numerical Oracle

The language-model layer enters the decision process only after deterministic evidence has been constructed. Its role is analogous to an investment-committee secretary or senior synthesizer. It receives the Buyer Mandate and compact candidate packets and is asked to identify leading opportunities, explain trade-offs, state principal concerns and surface unresolved conflicts.

This bounded role has three advantages.

First, it reduces hallucination risk. The model is not asked to invent property facts or search an uncontrolled world. It reasons over supplied evidence.

Second, it makes errors easier to diagnose. If the monthly payment is wrong, the financial engine can be inspected. If the mandate misinterprets the buyer, the semantic parser can be inspected. If the synthesis overstates a candidate's advantages, the deliberation output can be compared with the packet.

Third, it preserves substitutability. The language model can be changed without rewriting the financial engine or governance layer. Conversely, the data adapter can be replaced without redesigning the semantic architecture.

Image omitted from the lightweight GitHub edition.

Figure 8: Claude Sonnet 5 committee deliberation — Code Unit 7 places the language model at the committee-synthesis layer. Claude Sonnet 5 receives the Buyer Mandate and deterministic candidate packets, produces a bounded shortlist, states the case for each opportunity, identifies its principal concern, and surfaces unresolved mandate conflicts without replacing the underlying calculations or governance.

This architecture is therefore neither “LLM-only” nor “rules-only.” It is a layered system in which different forms of intelligence are deliberately assigned to different tasks.

Robustness: Does the Recommendation Survive a Different World?

A recommendation is conditional on assumptions. Real-estate affordability is particularly sensitive to interest rates, so the notebook reruns the financial engine and complete deterministic swarm across alternative mortgage-rate scenarios.

Let the ranking under mortgage rate r be

$$\pi(r) = \text{rank}\{S_j(r)\}.$$

A robust recommendation should not be understood merely by its baseline rank. We also care about rank stability:

$$\Delta_j(r_1, r_2) = |\text{rank}_j(r_1) - \text{rank}_j(r_2)|.$$

Large changes in Δ_j reveal candidates whose attractiveness depends heavily on financing assumptions. A property may be excellent at a low rate and infeasible at a high rate. The correct system response is not to hide this instability but to expose it.

Image omitted from the lightweight GitHub edition.

Figure 9: Interest-rate sensitivity and recommendation robustness — Code Unit 8 reruns financing and the deterministic 50-agent evaluation at mortgage rates of 4.5, 5.5, 6.5, 7.5 and 8.5 percent. The exercise asks whether the leading opportunity remains attractive as the financing environment changes, turning robustness into an observable property of the recommendation.

The same architecture can support many other stress dimensions: down payment, insurance shocks, renovation overruns, holding horizon, appreciation assumptions, family-size changes, commute changes, tax changes and climate scenarios.

A useful future metric is *recommendation resilience*: the proportion of plausible scenarios under which a candidate remains feasible and Pareto-efficient. Such a metric would not eliminate uncertainty. It would make the uncertainty operational.

From Research Architecture to an Executable Decision Pipeline

By Code Unit 9, the notebook has all required components but still needs an operational contract. The function `run_swarm` closes the architecture into a callable pipeline.

Conceptually, the sequence is:

Conversation → Buyer Mandate → Market → Financial State → 50 Specialists Coalitions → Governance → Pareto Set → Deliberation → Recommendation

The function returns five classes of output: a narrative synthesis, a structured shortlist, a coalition visualization, the formal Buyer Mandate and model provenance. This is significant because explainability is not confined to prose. The user can inspect what the system believed, what it calculated, how coalitions differed and whether live language reasoning or fallback logic was used.

Image omitted from the lightweight GitHub edition.

Figure 10: The complete end-to-end swarm pipeline — Code Unit 9 assembles the experiment into the callable `run_swarm` pipeline: conversation interpretation, financial enrichment, 50-agent evaluation, governance, Pareto analysis, candidate packets, committee deliberation, formatted shortlist, coalition visualization and provenance.

Operational provenance is especially important in hybrid systems. If an API call fails and deterministic fallback is used, that fact should be visible. A recommendation generated with live semantic interpretation is not operationally identical to one generated entirely by fallback rules, even if the final shortlist happens to be similar.

The Colab Application: Why the Demo Matters

The final notebook stage converts the architecture into a Gradio application that runs inside Google Colab. This may appear to be merely a presentation layer, but it is analytically important. An executable interface forces the research design to confront the complete user journey. Figure 11 should therefore be read as the visualization of *Code Unit 10 itself*: the construction of the interactive Gradio application. A separate image, Figure 12, is reserved for the end of the paper because it visualizes the finished application in an iPhone rather than describing another notebook cell.

The app begins with a conversation rather than a filter wall. This is not cosmetic. It reflects the thesis that the system's first task is to understand the decision before searching the opportunity space.

The client then receives an executive synthesis, a governed shortlist and a coalition comparison. The structured Buyer Mandate and model provenance are available as a transparency layer rather than being placed at the center of the consumer experience. The result resembles a digital investment committee: the buyer sees the recommendation while retaining the ability to inspect the machine.

Image omitted from the lightweight GitHub edition.

Figure 11: The interactive Colab application — Code Unit 10 wraps the complete pipeline in a Gradio interface. The buyer supplies a natural-language description of needs, chooses how many opportunities to inspect, and receives the committee narrative, governed shortlist, coalition comparison, structured Buyer Mandate and model provenance in one executable Colab application.

The demo is intentionally pedagogical. It lacks many components required for a production real-estate product: live listing ingestion, identity and access management, geospatial routing, comparable-sales analysis, document review, inspection workflows, image analysis, professional licensing controls, jurisdiction-specific disclosures, security hardening, privacy engineering, fair-housing controls and systematic empirical validation.

Yet the demo is powerful precisely because it is complete enough to reveal the architecture. The user can move from an ordinary sentence to a structured mandate, through a 50-agent committee, into an explainable shortlist. The notebook therefore functions as an executable research argument.

Why the Architecture Can Be More Resilient Than a Monolithic Recommender

The term *resilience* should be used carefully. The notebook does not empirically prove that the swarm is more accurate than every monolithic model. It does, however, instantiate several mechanisms that can make failure more observable and less catastrophic.

Functional decomposition

Errors are localized. Semantic interpretation, financial mathematics, scoring, governance, Pareto analysis and synthesis are separate stages. A failure in one layer does not automatically corrupt every other layer invisibly.

Diversity of objective functions

Specialists are intentionally biased toward different dimensions. In a monolithic score, a modeling error can dominate the complete recommendation. In a heterogeneous committee, a property that looks attractive to one coalition can still be challenged by another.

Preserved dissent

Dispersion is recorded. This creates a signal of epistemic fragility. High disagreement can trigger human review, additional data collection or a more conservative recommendation.

External governance

Hard constraints are not entrusted to generative reasoning. This reduces the risk that a persuasive narrative overrides feasibility.

Graceful degradation

The notebook includes deterministic fallbacks when the live language-model layer is unavailable. This is an architectural form of fault tolerance. The experience becomes less semantically rich, but the complete pipeline need not disappear.

Scenario sensitivity

The system explicitly reruns itself under alternative interest rates. A recommendation that survives multiple plausible states deserves a different interpretation from one that depends on a narrow baseline.

RESILIENCE AS AN ARCHITECTURAL PROPERTY

In this prototype, resilience does not mean that the swarm cannot be wrong. It means that the system contains multiple opportunities to detect, isolate, challenge and disclose error: specialist disagreement, governance gates, deterministic arithmetic, Pareto alternatives, scenario tests, provenance and human-readable explanations.

The Most Important Extension: Revealed Preference

The initial Buyer Mandate is only a hypothesis about the buyer. A mature system should learn from behavior.

Suppose the buyer repeatedly rejects homes with long commutes even though commute received only moderate stated importance. Suppose the buyer saves older properties despite declaring a strong preference for new construction. Suppose school quality was described as paramount, but every saved home lies in a highly walkable urban neighborhood.

These actions contain information about latent utility.

Let M_t denote the mandate after interaction t and let a_t represent an observed buyer action such as save, reject, tour, revisit or request-more-information. A learning system could update the mandate according to

$$M_{t+1} = \mathcal{U}(M_t, a_t, \mathbf{x}_{j_t}),$$

where $\mathcal{U}$ is a preference-learning operator.

The conceptual loop becomes

Stated Preference → Observed Choice → Revealed Preference → Revised Mandate → New Search

This extension changes the nature of the product. The system is no longer merely a property recommender. It becomes an adaptive buyer's advisory committee that learns what the client values while keeping the evolution of that belief inspectable.

Revealed-preference learning also creates new governance questions. Behavioral data can be noisy. A saved property may reflect curiosity rather than utility. Repeated browsing can be influenced by presentation order. The system should therefore update preferences gradually and allow the buyer to inspect and correct inferred changes.

Pedagogical Value of the Experiment

The notebook is especially useful as a teaching artifact because it makes several abstract AI concepts concrete.

First, it shows that *agents* are an organizational design problem. The difficult question is not how to create fifty model calls. It is how to assign differentiated mandates, define evidence boundaries, aggregate outputs and govern the result.

Second, it demonstrates that *autonomy* should be decomposed. The system can be autonomous in mandate parsing and committee synthesis while remaining deterministic in arithmetic and constraints.

Third, it shows that *explainability* is architectural. A final paragraph explaining a black-box score is weaker than a system that preserves the mandate, specialist views, coalition scores, violations, Pareto status, scenario sensitivity and provenance.

Fourth, it demonstrates that *governance* is not a post-processing disclaimer. Governance is embedded in the computational path.

Fifth, the application demonstrates how research code can become a decision interface. This is crucial for executives and financial practitioners. The value of AI architecture becomes much clearer when the chain from abstract design to user experience is visible.

Limitations

The experiment has important limitations.

The first is synthetic data. Although the generated market preserves useful correlations, it does not reproduce the legal, geographic, informational and behavioral complexity of an actual housing market. Results should therefore be interpreted as architectural demonstrations, not empirical claims about real property performance.

The second is handcrafted scoring. Specialist feature maps and coalition weights are transparent, which is valuable pedagogically, but they are not estimated from observed buyer welfare or transaction outcomes. A production system would require calibration and validation.

The third is limited independence among agents. The deterministic specialists share a common feature-engineering foundation, and the language-model components share a common model family. Their errors may therefore be correlated. True cognitive diversity may require different models, tools, data sources, algorithms or independently trained subsystems.

The fourth is the absence of transaction frictions and legal detail. Inspection findings, title issues, financing contingencies, taxes, insurance exclusions, zoning, condominium documents, local regulations and broker practices can materially alter a real purchase.

The fifth is fairness. Variables such as neighborhood, schools and safety can interact with protected characteristics and historical patterns of segregation. A production system would require careful fair-housing review, data governance and restrictions on the use of proxies. The objective should be to support lawful buyer objectives without reproducing discriminatory steering.

The sixth is evaluation. A compelling demo is not evidence of superior decision quality. The architecture must be compared against meaningful baselines.

A Research Agenda for Empirical Validation

The prototype creates a platform for a more rigorous experimental program.

Ablation tests

Remove one architectural component at a time: governance, Pareto analysis, dissent, coalition structure, language-model synthesis or revealed-preference updating. Measure how rankings, feasibility violations and explanation quality change.

Swarm-size convergence

Run the architecture with 5, 10, 20, 30 and 50 agents. If rankings stabilize at a much smaller swarm, the additional agents may not justify their computational cost. If specialist coverage rather than count drives performance, this should become visible.

Monolithic versus swarm comparison

Give a single language model the same Buyer Mandate and candidate evidence and compare its recommendations with the hybrid swarm. The relevant outcomes include hard-constraint violations, ranking stability, calibration, explanation consistency and sensitivity to prompt perturbation.

Human expert comparison

Ask experienced buyer agents, mortgage professionals, property investors and household decision makers to evaluate standardized cases. Compare the system's opportunity sets and explanations with expert panels rather than assuming that an AI baseline is sufficient.

Revealed-preference prediction

After observing a sequence of buyer choices, test whether the adaptive mandate predicts later saves, tours or final choices better than the original interview mandate.

Scenario robustness

Vary rates, insurance, taxes, renovation costs, holding horizon and commute assumptions. Measure rank stability and identify candidates whose attractiveness is fragile.

Data provenance and factuality

With live data, every property attribute should carry source, timestamp and confidence. The system should distinguish listing claims, public records, estimated values and model-derived quantities.

Outcome evaluation

The strongest long-run test would examine whether the architecture improves decision satisfaction, financial sustainability and ex post regret relative to simpler search tools. Such work would require careful study design and substantial time horizons.

A CRITICAL SCIENTIFIC DISTINCTION

The notebook demonstrates *architectural plausibility*, not *empirical superiority*. The next research phase should attempt to falsify the proposition that a heterogeneous governed swarm adds value. If a smaller or monolithic system performs equally well on robust outcome measures, the more complex architecture should not be preferred merely because it is more elaborate.

Toward a Production Architecture

A production system can preserve the conceptual structure while replacing experimental components.

The synthetic adapter would become a governed data layer consuming authorized listing feeds, property records, geospatial data, mortgage rates, tax information, insurance data, climate information and other permitted sources. Each field would carry provenance and freshness.

The conversational interview would become multi-turn. Rather than extracting all preferences from one paragraph, the system would ask high-information questions when uncertainty is material. If budget and commute create an infeasible search, the system should ask which constraint is more negotiable.

The financial engine would become jurisdiction-aware and include transaction costs, mortgage products, taxes, insurance and scenario-specific cash flows.

The specialist society could become genuinely heterogeneous computationally. Some agents might be deterministic models, others retrieval agents, geospatial tools, computer-vision models, document agents or language-model specialists. A home-inspection agent need not share the same architecture as a mortgage agent.

The governance layer would expand to include privacy, security, fair-housing restrictions, licensing boundaries, source quality, conflict-of-interest controls and escalation to human professionals.

Finally, the user interface would evolve from a one-session Colab app into a persistent decision workspace containing maps, saved properties, rejected properties, visit notes, documents, financing scenarios and the evolving Buyer Mandate.

The important point is that the notebook already contains the skeleton of this production system. The pedagogical prototype is not production-ready, but its decomposition is extensible.

Discussion: What the Experiment Teaches About Swarms

The most important lesson is that a swarm is not valuable because fifty is a large number. The value comes from *structured plurality*. The architecture creates multiple views, preserves their boundaries, gives them a common evidence substrate and then subjects their collective output to governance.

This resembles the institutional design of sophisticated human organizations. A bank does not ask the tax department to become the credit department, or the risk function to disappear because the investment team is enthusiastic. Different functions exist because complex decisions contain different failure modes. The swarm translates that organizational principle into software.

The second lesson is that hybrid architectures can reduce unnecessary epistemic ambiguity. Exact formulas remain exact. Language models are used where language is the problem. Governance is explicit. This makes the system easier to audit and teach.

The third lesson is that disagreement can be a feature. Much of machine learning is designed to produce a single prediction. Complex decisions often require something else: a map of the disagreement. A family may rationally accept a weaker investment profile for a better school environment. Another buyer may do the reverse. The system's responsibility is not to pretend those value judgments have disappeared.

The fourth lesson is that the user interface is part of the epistemology. A filter-based interface implies that the buyer already knows the correct variables and thresholds. A conversational interface allows the system to help construct the decision itself. The Buyer Mandate becomes the bridge between human ambiguity and computational precision.

The fifth lesson is that a working demo has research value. By forcing the architecture into an executable application, hidden assumptions become visible. What exactly enters the system? What happens when the API is absent? Which outputs are shown? Which are hidden? How are constraints represented? How does a user inspect provenance? These questions are easy to postpone in a conceptual paper and impossible to avoid in a working app.

The architecture in the hand of the buyer

The final visualization is intentionally placed here, near the end of the argument rather than in the technical sequence of notebook cells. Figure 12 is not another computational stage. It is a visualization of the completed application as it could appear on an iPhone. Its role is therefore different from Figures 2–11: those figures explain how the system is built; this one shows what the completed research architecture means from the user’s point of view.

That distinction is central to the paper. The buyer does not need to experience fifty agents, coalition weights, Pareto dominance, structured JSON or fallback logic as separate pieces of machinery. Those mechanisms belong behind the interface. What should reach the user is a coherent advisory experience: describe the household and objectives in ordinary language, receive a small set of defensible opportunities, understand the principal trade-offs, and retain the ability to inspect the mandate and evidence when desired. The mobile visualization therefore closes the conceptual loop from institutional architecture back to human decision making.

Image omitted from the lightweight GitHub edition.

Figure 12: The completed real-estate swarm as a mobile decision experience — Unlike Figures 1–10, which document the notebook architecture and its ten code units, this final image visualizes how the resulting application can look in an iPhone form factor. It is placed near the end of the paper because it represents the user-facing culmination of the research design rather than an additional computational stage.

Conclusion

Residential real estate is a compelling laboratory for multi-agent decision intelligence because it combines financial engineering, household utility, location, subjective preference, long horizons, uncertainty and large irreversible commitments. The apparent simplicity of online property search conceals a much richer problem: the buyer must discover not only which homes exist, but what he or she actually values, which trade-offs are acceptable, which constraints are binding and how robust a choice remains when assumptions change.

The 50-agent prototype developed in the companion notebook addresses this complexity by refusing to collapse the problem too early. A conversational layer constructs a Buyer Mandate. A synthetic but structured market provides a controlled opportunity space. A deterministic financial engine calculates carrying costs. Fifty heterogeneous specialists examine the properties through Finance, Investment, Family, Location, Lifestyle, Risk and Governance lenses. Coalition aggregation creates collective evidence without erasing dissent. Governance prevents hard constraints from becoming mere preferences. Pareto analysis preserves efficient alternatives. A language model synthesizes bounded factual packets rather than inventing the numerical state. Sensitivity analysis tests whether the recommendation survives different financing regimes. Provenance records how the result was produced. Finally, a Gradio application converts the complete architecture into something a user can touch.

The prototype is intentionally pedagogical. It can be improved substantially, and many of its components should be replaced before any production use. But this limitation should not obscure what the experiment demonstrates. A swarm can be more than a theatrical collection of model personas. It can be designed as an artificial institution: a set of bounded specialists with differentiated responsibilities, shared evidence, explicit aggregation, preserved disagreement, deterministic controls and a governed path to action.

The deeper research opportunity lies in making that institution adaptive. Once the buyer begins saving, rejecting, touring and revisiting properties, behavior becomes evidence about latent preferences. The Buyer Mandate can evolve. The swarm can search again. The system can become less a recommender and more a persistent artificial advisory committee.

The resulting research program is broader than real estate. Many consequential decisions share the same structure: the objective is incompletely specified, evidence is heterogeneous, specialists disagree, some constraints are non-negotiable, assumptions change and the final recommendation must remain explainable. Real estate provides an unusually intuitive demonstration of how swarms and hybrid architectures can address that class of problem.

The scientific claim should remain modest until empirical tests are performed. We have not shown that fifty agents are always better than one. We have shown something more foundational: it is possible to organize distributed artificial intelligence so that complexity is decomposed rather than hidden, disagreement is retained rather than averaged away, exact computation remains exact, language reasoning is bounded by evidence, governance is computational rather than rhetorical, and the complete process can be delivered through a usable application.

That is sufficient to turn a demonstration into a research agenda.

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COMPANION COMPUTATIONAL ARTIFACT

The executable research record for this paper is the companion notebook NB_50_AGENT_REAL_ESTATE_SWARM_CLAUDE_SONNET_5_10_CELLS.ipynb. It contains the resilient Claude Sonnet 5 environment, synthetic 1,200-property market, replaceable data adapter, conversational Buyer Mandate, deterministic financial engine, fifty-agent taxonomy, specialist scoring system, governance gates, coalition aggregation, dissent measure, Pareto frontier, candidate evidence packets, committee-level language-model deliberation, interest-rate sensitivity experiment, end-to-end callable pipeline, provenance record and premium Gradio application described in this paper.

Notebook location: [Open the companion Google Colab notebook.](#)